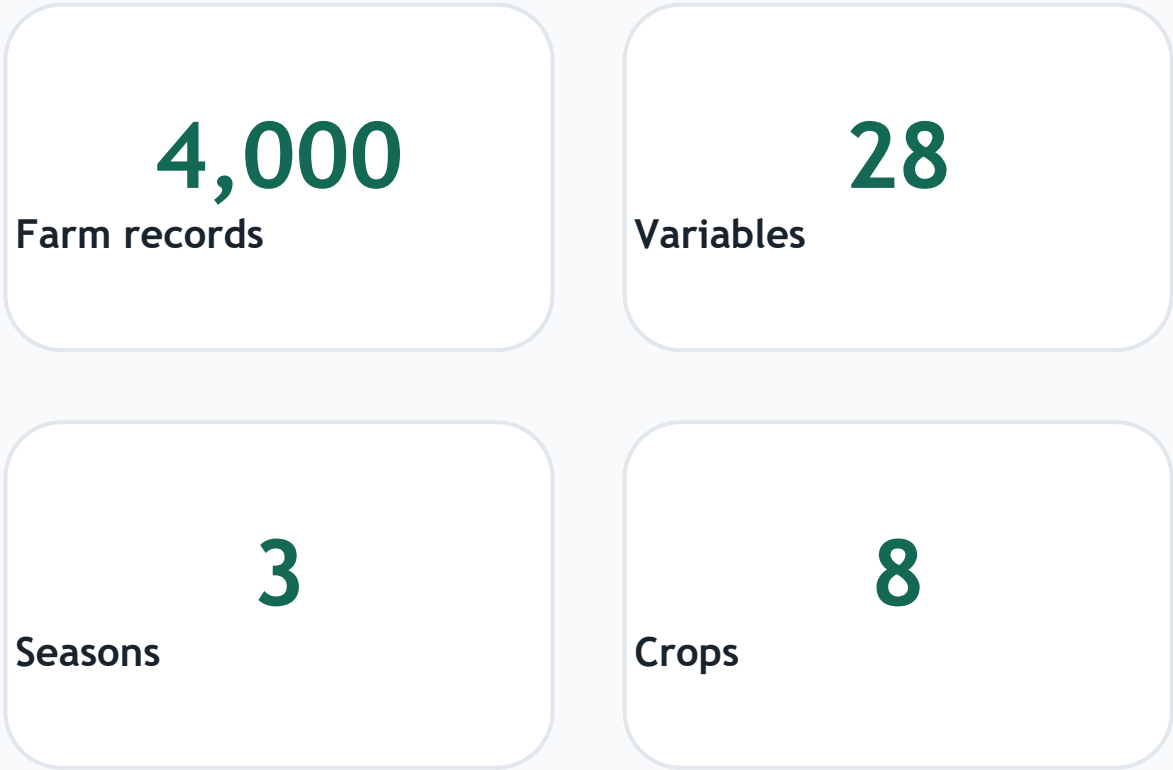


Seasonal Agriculture Performance Analysis

VOIS AICTE Major Project — Data Visualization / Data Analytics

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Problem Statement

- Agricultural performance changes with seasonal environmental conditions, farming practices, resource availability and market conditions.
- Raw agricultural data does not directly show how yield, production, resource use and economic performance vary between Kharif, Rabi and Zaid.
- The project analyzes seasonal differences, trends, relationships and variations using descriptive statistics, visualization and statistical testing.
- The goal is to turn farm-level records into evidence-based insights for seasonal agricultural planning.

Project Description

- Dataset contains 4,000 farm records across 8 states, 8 crops and 3 agricultural seasons.
- Variables cover farm area, rainfall, temperature, humidity, sunlight, soil conditions, nutrients, irrigation, fertilizer, pesticides, seed quality, yield and production.
- Economic variables include market price, total cost, revenue and profit; resource variables include water use and water efficiency.
- Analysis focuses on seasonal performance and also compares crop, irrigation and state-level patterns.
- Missing values were identified and handled transparently; exact duplicates were checked before analysis.

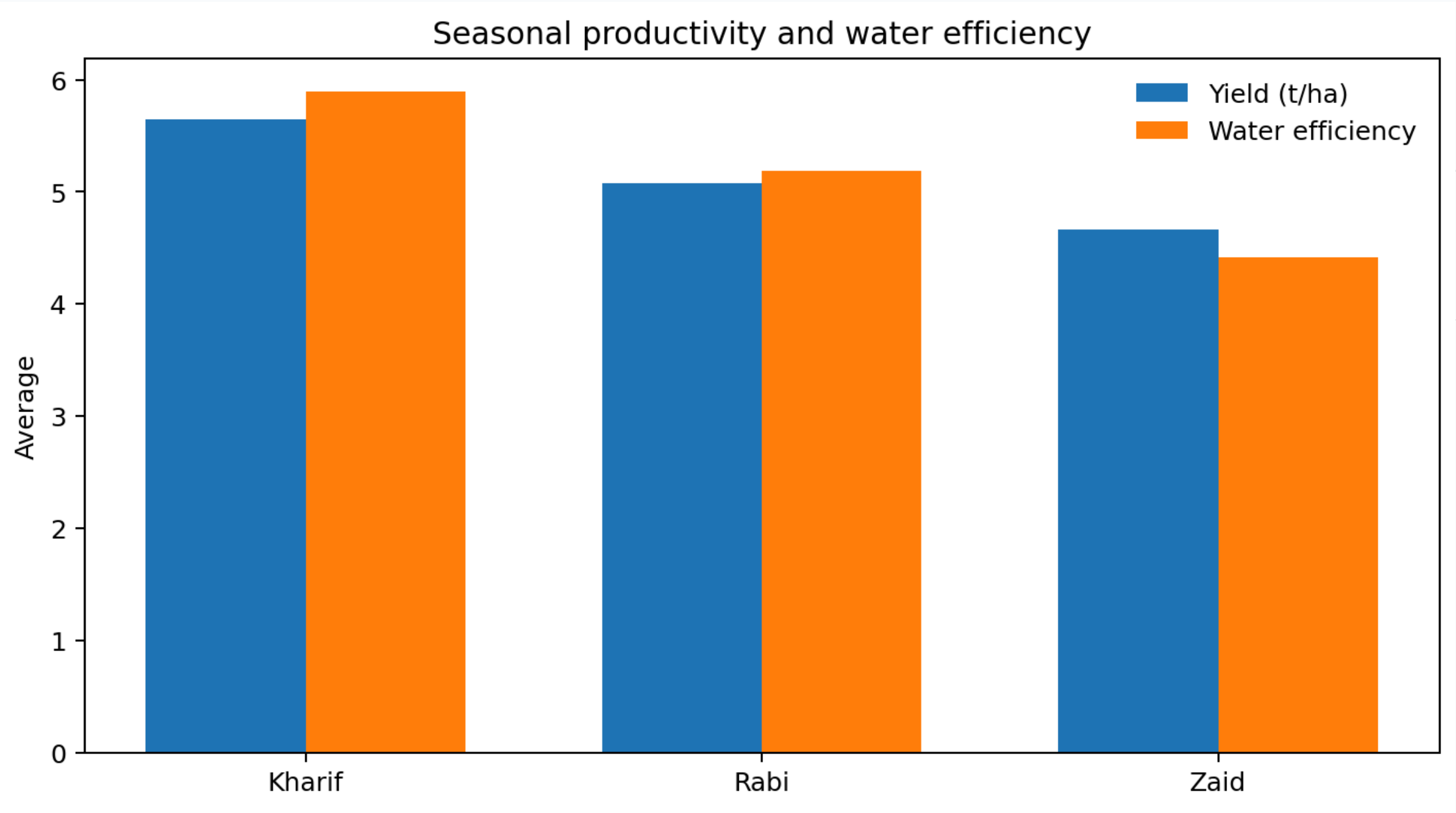
Who Are the End Users?

- Farmers — compare crop and seasonal profitability and resource requirements.
- Agricultural planners — identify seasons and regions requiring additional attention.
- Irrigation/resource managers — compare water use and efficiency across irrigation methods.
- Agribusiness stakeholders — understand revenue, cost and market-related performance.
- Data analysts/researchers — use the dataset for deeper statistical and predictive studies.

Technology Used

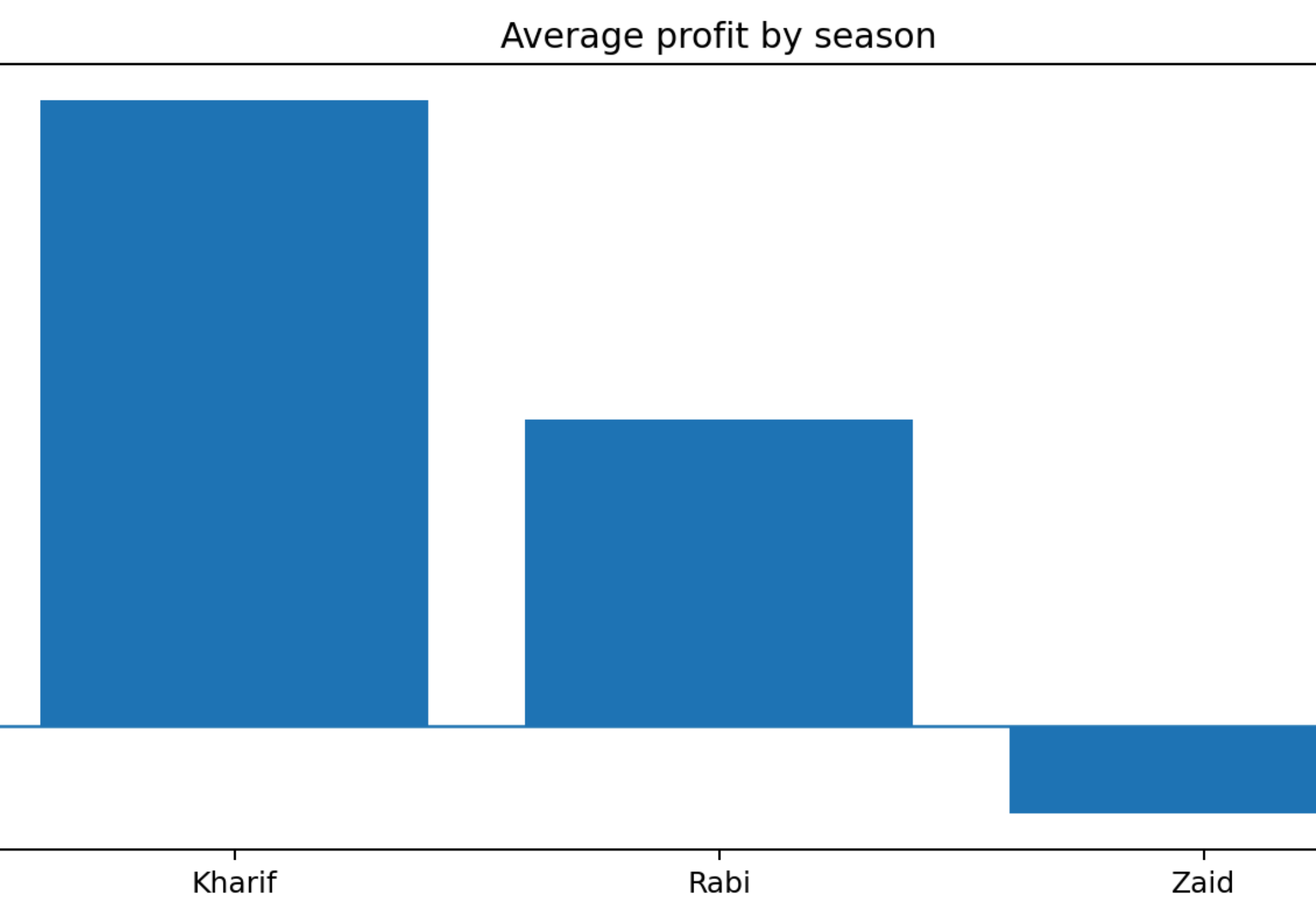
- Python — core programming and analysis
- Pandas — data loading, cleaning, grouping and aggregation
- NumPy — numerical operations
- Matplotlib — data visualization
- SciPy — statistical testing / one-way ANOVA
- Jupyter Notebook — reproducible analysis and documentation
- PowerPoint — presentation of analytical findings

RESULTS – Seasonal Productivity



- Kharif has the highest average yield: 5.64 t/ha.
- Rabi averages 5.08 t/ha, while Zaid is lowest at 4.67 t/ha.
- Water efficiency also decreases from Kharif (5.89) to Zaid (4.41 t/1000 m³).
- Seasonal patterns should be interpreted together with crop mix, irrigation and environmental conditions.

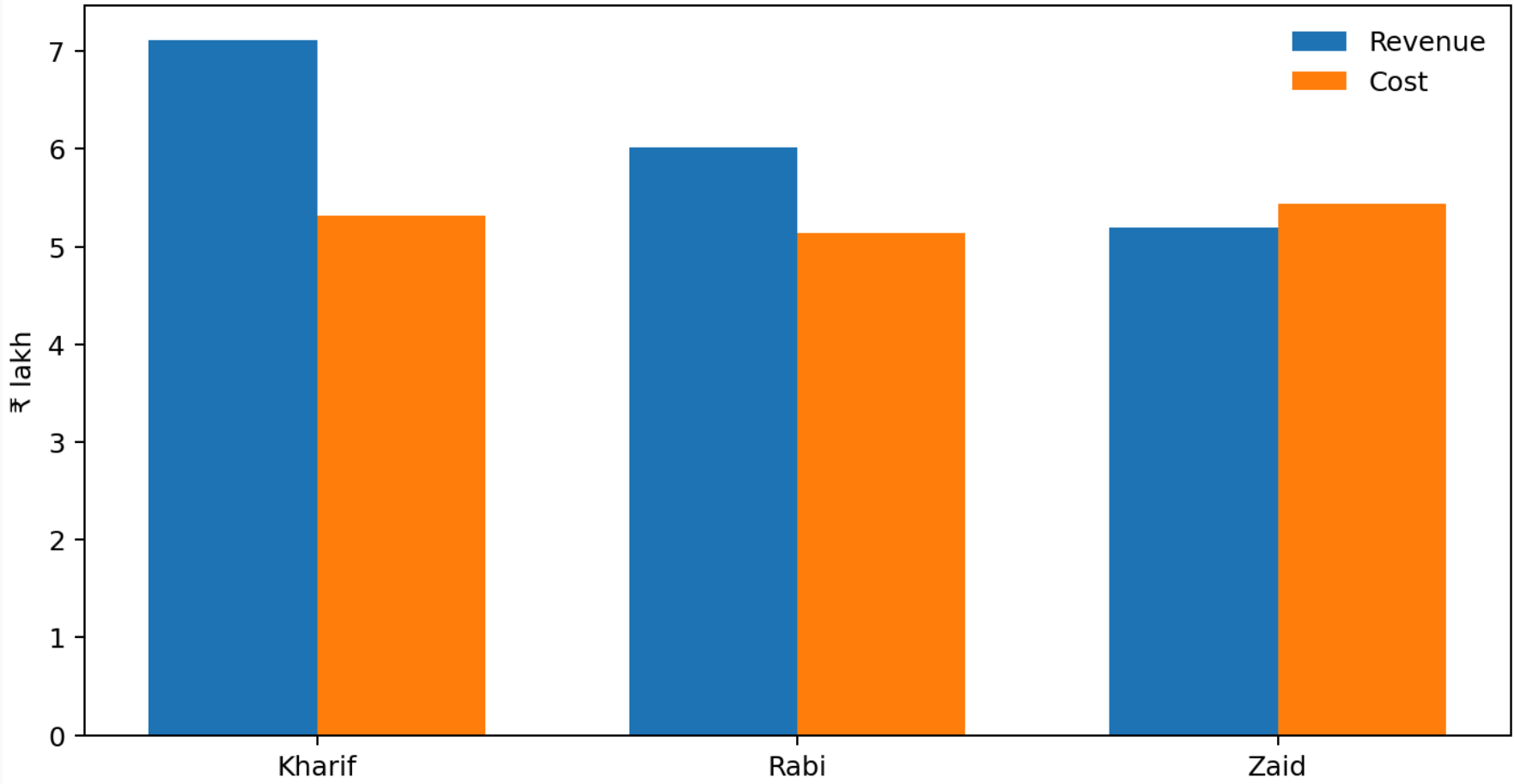
RESULTS – Profitability



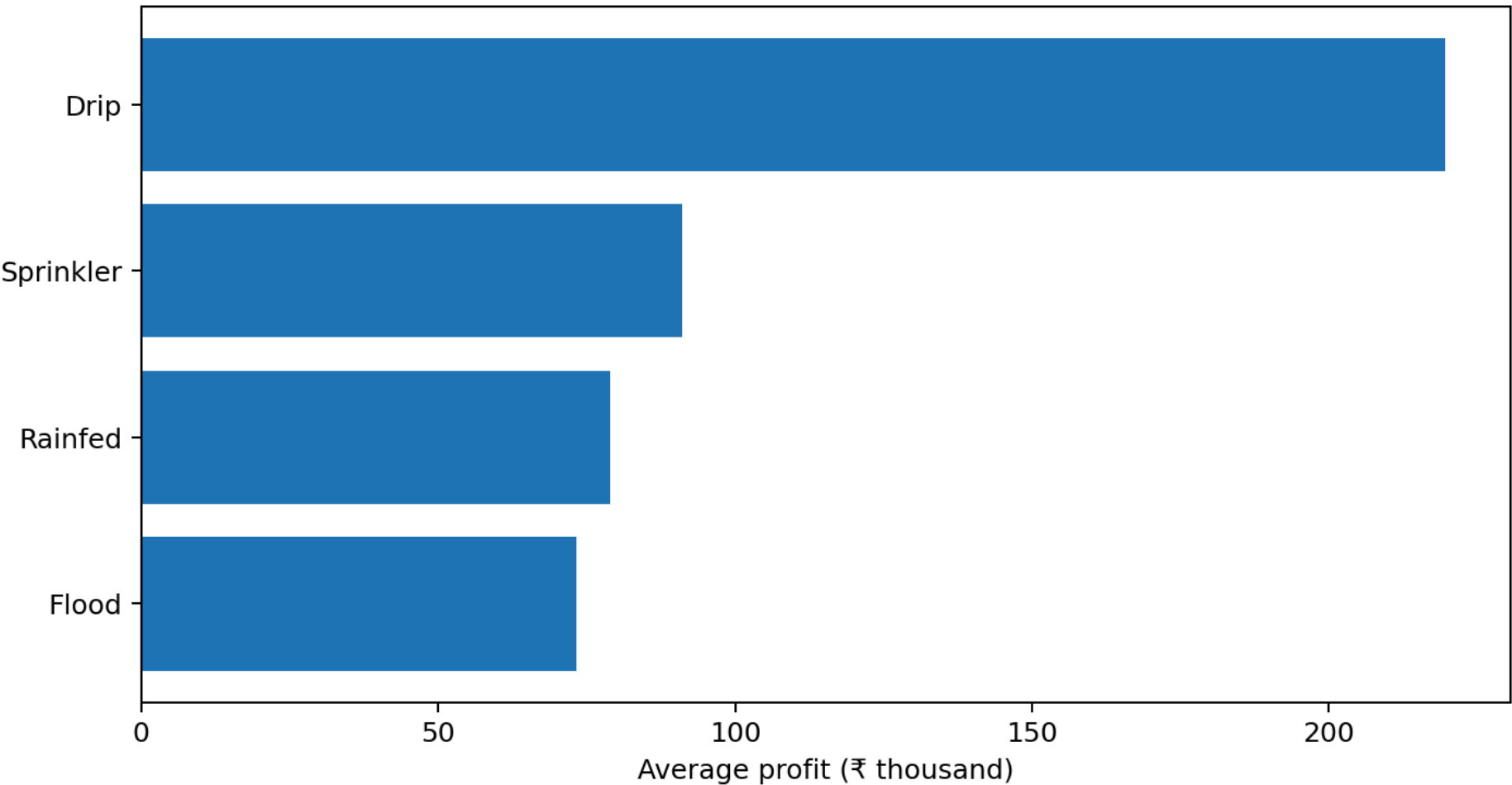
- Kharif average profit: ₹179k; 57.8% of farms are profitable.
- Rabi average profit: ₹88k; 48.9% profitable.
- Zaid average profit: ₹-25k; only 35.5% profitable.
- ANOVA: $F=34.29$, $p=1.71e-15 \rightarrow$ significant seasonal difference in mean profit.

RESULTS – Revenue, Cost & Irrigation

Average revenue vs cost

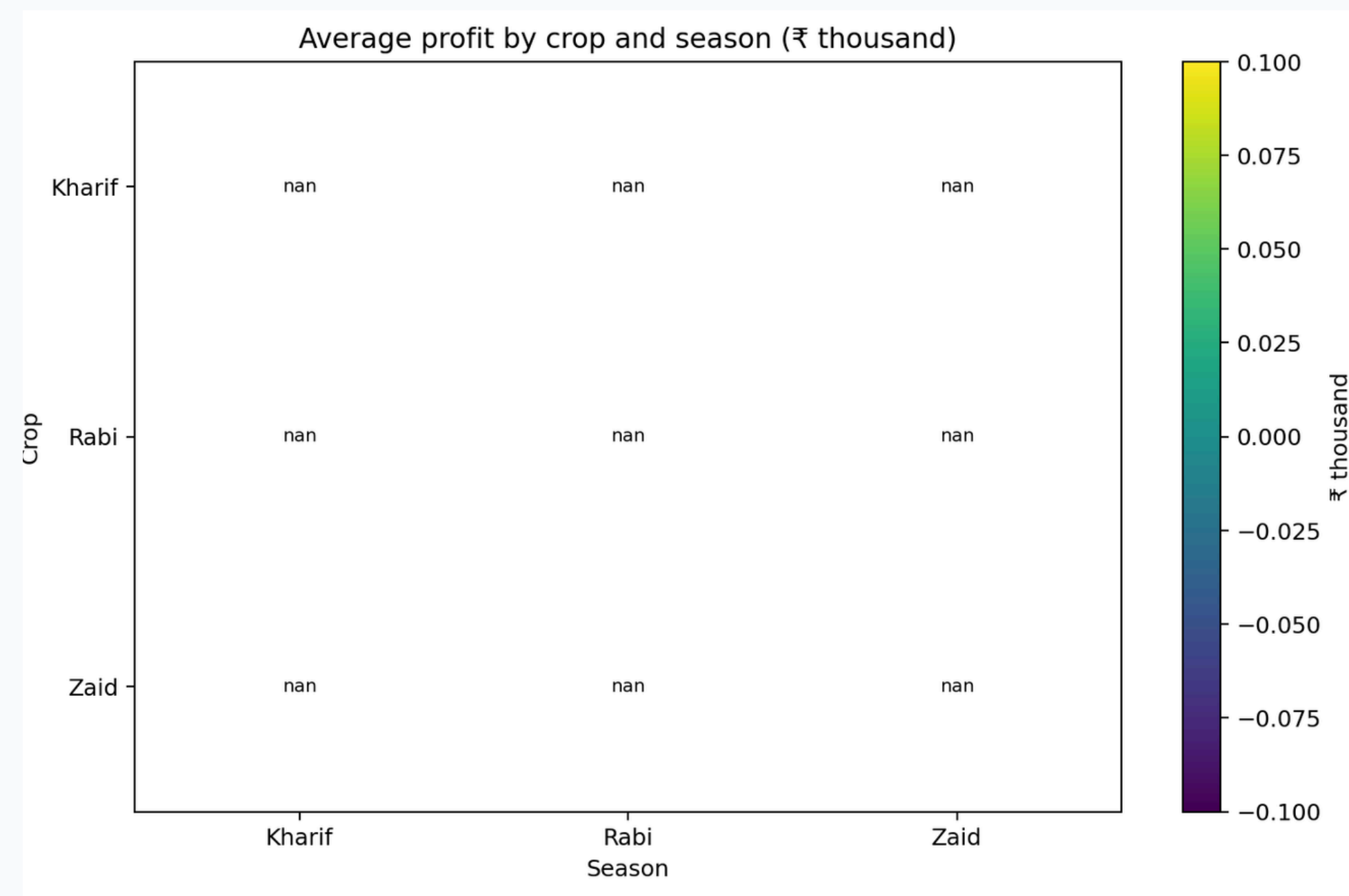


Profit by irrigation method



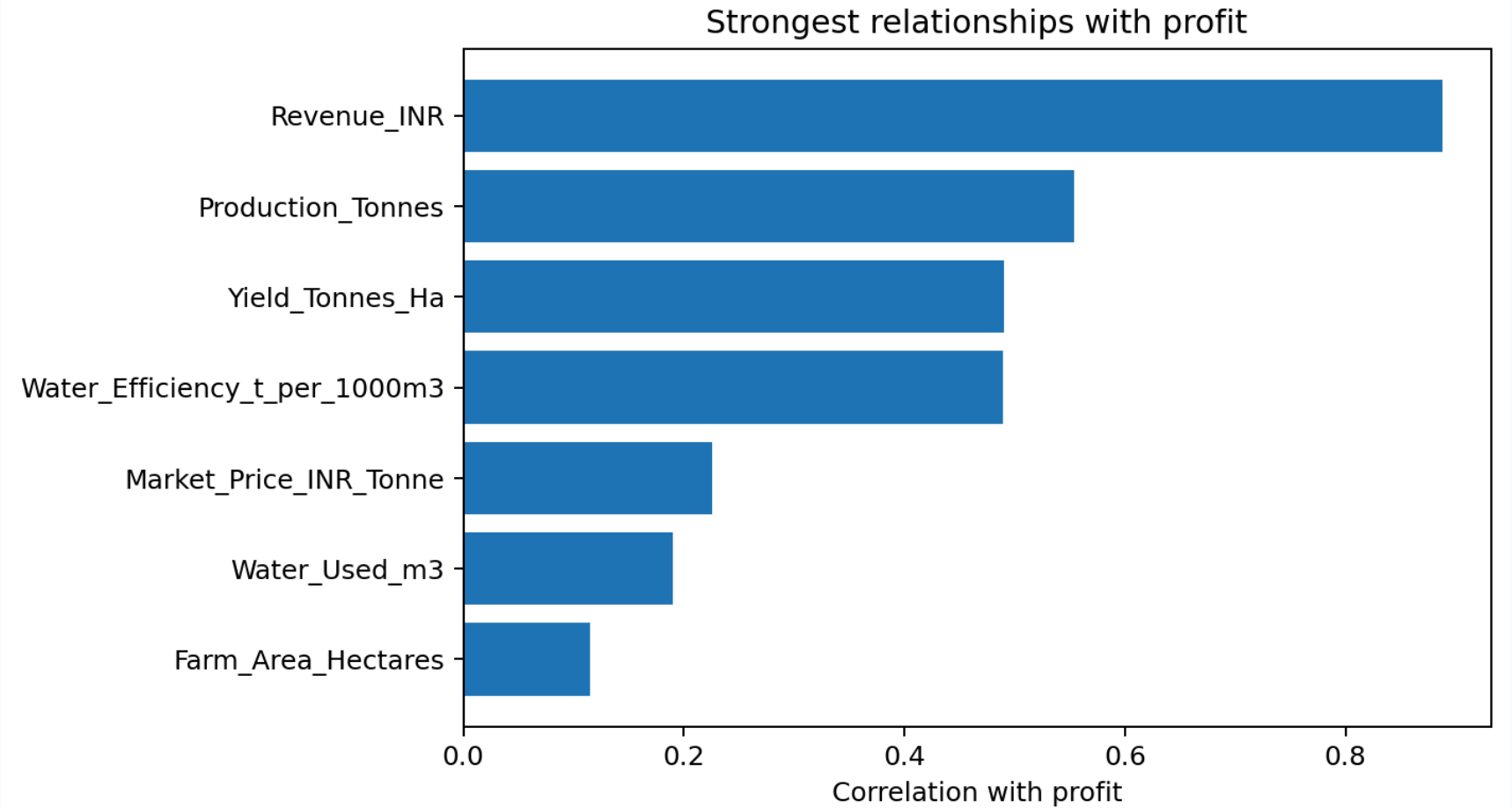
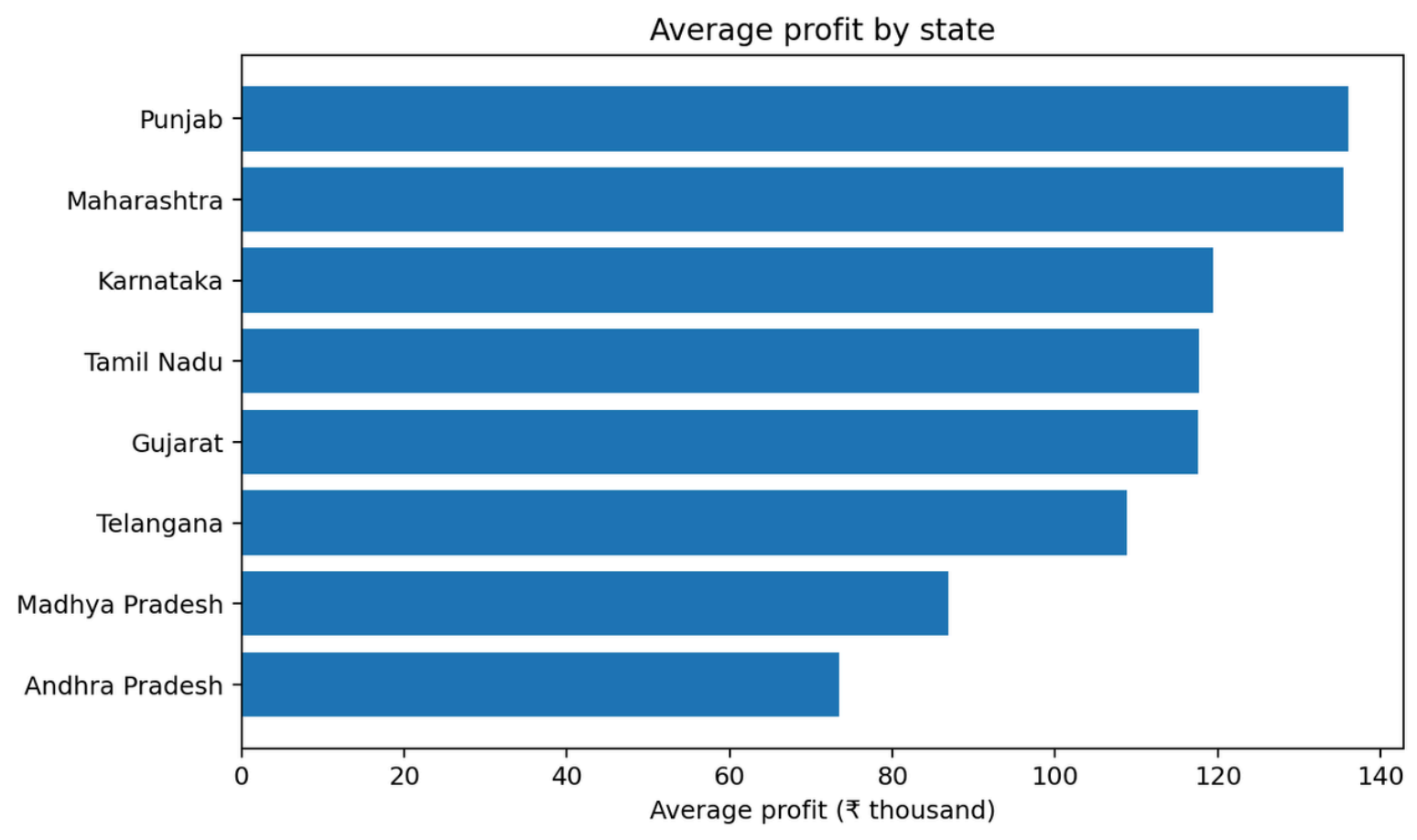
- Zaid has the highest average cost (₹5.44 lakh) and lowest average revenue (₹5.19 lakh).
- Drip has the highest average profit among irrigation methods: ₹220k.
- Flood uses the most water: 8,026 m³ on average, with the lowest water efficiency (3.44).

RESULTS – Crop × Season



- Highest average crop-season profit: Sugarcane in Kharif (₹1,001k).
- Lowest: Rice in Zaid (₹-227k).
- Sugarcane and Chilli are consistently strong profit categories in this dataset.
- Wheat, Rice and Maize show repeated negative average profit across seasons.
- Planning should therefore be crop-specific, not based on season alone.

RESULTS – Regional & Relationship Analysis



- Punjab has the highest state average profit: ₹136k.
- Andhra Pradesh has the lowest among the listed states: ₹73k.
- Revenue has the strongest correlation with profit ($r=0.887$); production and yield are also positively associated.
- Correlation indicates association, not causation.

Future Scope

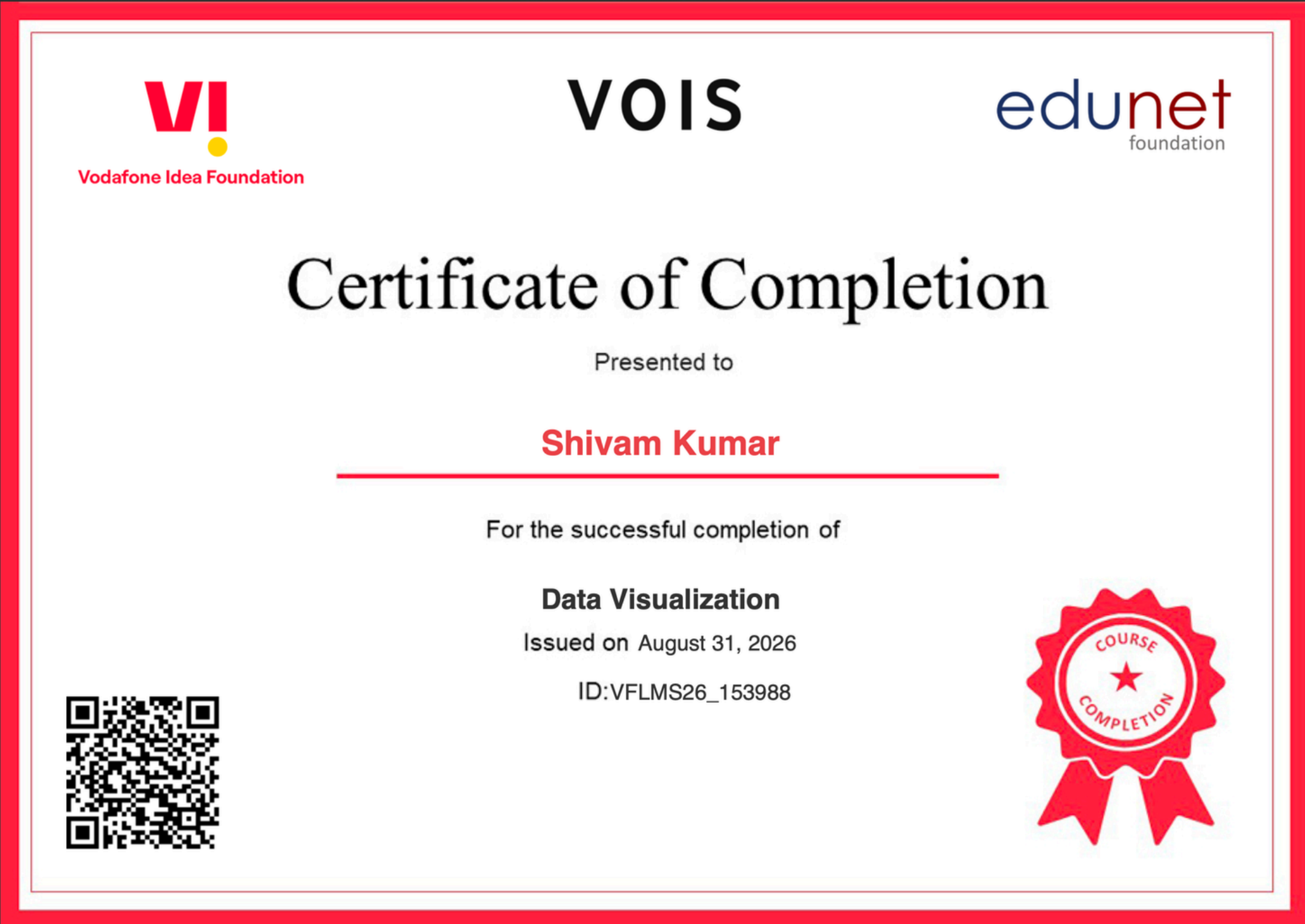
- Build a multivariate regression model to predict profit or yield.
- Control for crop, state, farm area, irrigation, climate and input variables to isolate relationships.
- Develop district-level dashboards for localized agricultural planning.
- Add time-series data to study trends across years rather than only seasonal cross-sections.
- Evaluate irrigation adoption using cost-benefit analysis and water-saving metrics.
- Explore predictive risk models for disease/pest risk and seasonal underperformance.

GitHub Link

Project Notebook / Source Code

GitHub: <https://github.com/user-23f2000797/Seasonal-Agriculture-Performance-Analysis>

VOIS Course Completion Certificate



Thank You

Seasonal Agriculture Performance Analysis